

SMOOTHCONV AND DUPLEXCONV: COMPLEMENTARY MANDARIN MULTI-PARTY CONVERSATIONAL SPEECH CORPORA FOR SPEECH INTERACTION

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ABSTRACT

Recent advances in large audio language models (LALMs) have driven the development of natural and intelligent speech interaction systems. Such systems need to model complex conversational behaviors, including turn-taking, overlapping speech, and speaker coordination. Multi-party conversations provide a realistic setting for studying these behaviors, yet existing Mandarin conversational corpora often lack synchronized participant-level speech tracks and comprehensive annotations. In this work, we introduce **SmoothConv** and **DuplexConv**, two complementary Mandarin multi-party conversational speech corpora totaling 2,100 hours. SmoothConv provides human-verified conversations for reliable analysis and evaluation, while DuplexConv offers large-scale automatically annotated conversations through a scalable pipeline for model training. Both corpora provide synchronized participant-level speech tracks and multi-dimensional fine-grained annotations. We further release the **SmoothConv Benchmark** and evaluate these resources on speech separation, multi-speaker automatic speech recognition (MSASR), and turn detection tasks. Experimental results demonstrate the utility of the proposed resources for multi-party speech interaction modeling. The datasets, benchmark, and related resources are publicly available.¹

Index Terms— Multi-party conversational speech, speech interaction, speech corpus, benchmark

1. INTRODUCTION

Recent advances in large audio language models (LALMs) have made natural, real-time, and coherent speech interaction an increasingly important area. Beyond recognizing speech content and speaker identity, interactive speech systems need to understand conversational behaviors, including turn-taking, response timing, and paralinguistic cues. Multi-party conversations provide a natural setting for studying such behaviors, as multiple speakers coordinate their speech through turn exchanges, overlaps, and evolving interaction states. However, existing conversational resources often lack either synchronized participant-level speech tracks or comprehensive annotations of interaction behaviors, limiting their applicability to realistic multi-party speech interaction.

High-quality multi-party conversational data requires two complementary components: synchronized speech signals from individual participants and fine-grained annotations describing conversa-

Table 1. Comparison with representative conversational corpora. Multi-Track, Turn, Para., and Attr. denote synchronized participant-level speech tracks, turn-state annotations, paralinguistic information annotations, and speaker attributes, respectively. ✓/✗/△ indicate full, no, and partial coverage.

Corpora	Hours	Spk.	Multi-Track	Turn	Para.	Attr.
MagicData-RAMC [7]	180	2	✗	✗	✗	✗
Switchboard [1]	260	2	✓	△	✗	✗
Fisher [2]	2,000	2	✓	✗	✗	✗
CHiME-5 [3]	50	4	△	✗	✗	✗
MLC-SLM [8]	1,502	2	✗	✗	✗	✗
CANDOR [9]	850	2	✗	✓	✓	△
AMI [4]	100	4	✓	✓	△	✗
AliMeeting [5]	118	2–4	✗	✓	△	✗
AISHELL-4 [6]	120	4–8	✗	✗	✗	✗
SmoothConv	100	2–4	✓	✓	✓	✓
DuplexConv	2,000	2–4	✓	✓	✓	✓

tional content, speaker information, and interaction behaviors. Existing conversational corpora vary in their recording configurations and annotation schemes. Switchboard [1] and Fisher [2] contain natural conversations with separate speaker channels, but are limited to English two-speaker telephone conversations. CHiME-5 [3] captures realistic multi-speaker conversations in everyday environments using multi-channel recordings, but mainly targets robust ASR and speech separation without comprehensive fine-grained annotations. AMI [4] and AliMeeting [5] provide multi-party meeting recordings with speaker annotations, while AISHELL-4 [6] introduces Mandarin multi-party meeting data. However, these resources either rely on microphone array recordings or lack synchronized participant-level speech tracks and detailed annotations of interaction behaviors. Consequently, publicly available Mandarin resources remain insufficient for modeling realistic multi-party conversations.

Another challenge lies in scaling multi-dimensional annotation to large conversational datasets. While many conversational corpora provide basic speech transcriptions and speaker information, fine-grained descriptions of interaction behaviors and speaker attributes remain limited. Manual annotation provides reliable and detailed labels, but is costly and difficult to extend to large-scale data. Automatic annotation improves scalability, yet may introduce errors in recognition, alignment, and semantic interpretation. This motivates the development of Mandarin conversational resources that combine

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¹<https://github.com/qualialabsAI/SmoothConv-DuplexConv>

carefully verified data for reliable analysis and automatically constructed data for scalable model training.

To address these limitations, we introduce **SmoothConv** and **DuplexConv**, two complementary Mandarin multi-party conversational speech corpora totaling 2,100 hours across Tutoring and Social Chat scenarios. Both corpora provide synchronized participant-level speech tracks and multi-dimensional annotations while serving different research purposes. SmoothConv contains 100 hours of naturally occurring conversations with detailed human verification, supporting fine-grained analysis and reliable evaluation. DuplexConv contains 2,000 hours of automatically annotated conversations generated through a scalable pipeline, supporting large-scale model training. We further release the **SmoothConv Benchmark**, including evaluation data and protocols for reproducible multi-party speech interaction research. We evaluate these resources on three representative tasks spanning different aspects of multi-party speech interaction, including speech separation for participant-level signal recovery, multi-speaker automatic speech recognition (MSASR) for speaker-aware content understanding, and turn detection for conversational behavior modeling. The results demonstrate the utility of these resources for training and evaluating models on multi-party conversational speech, highlighting the complementary value of large-scale automatically annotated data and high-quality human-verified annotations.

2. DATASET

SmoothConv and DuplexConv are built from synchronized multi-party conversations collected from real-world online interactions. Unlike scripted, predefined-topic, or role-playing recordings, these sessions preserve spontaneous conversational behaviors, including turn exchanges, interruptions, speech overlaps, and dynamic interaction patterns. All recordings were collected with informed consent and anonymized before release, with no personally identifiable information retained.

SmoothConv emphasizes annotation quality through human verification and supports reliable analysis and evaluation, whereas DuplexConv adopts a scalable automatic annotation pipeline to provide large-scale training data.

2.1. Construction Pipeline

As shown in Fig. 1, the pipeline consists of two stages: per-channel recognition and speaker labeling, followed by multi-dimensional annotation.

Per-channel recognition and speaker labeling. For each participant track, VAD first detects speech regions while preserving the original timeline synchronization. Multiple ASR systems [10, 11, 12, 13] independently generate transcripts, and ROVER [14] fusion combines their outputs into a unified recognition result. Dasheng [15] is then applied to remove non-speech segments. Pyannote diarization [16] and Qwen3-ForcedAligner-0.6B [17] further refine speaker attribution and utterance boundaries. Finally, 3D-Speaker clustering [18] provides consistent speaker labels across the complete conversation.

Multi-dimensional fine-grained annotation. Based on aligned transcripts and speaker labels, we construct annotations for conversational turn states, paralinguistic information, and speaker attributes. To evaluate the quality of automatic annotations, we

randomly sampled a 20-hour subset from DuplexConv for human verification. The sampled data were independently reviewed by the same annotator team involved in SmoothConv annotation. A sample is considered valid when the automatically generated annotations satisfy the predefined quality criteria covering transcription, speaker labeling, and conversational interaction annotations. The human verification pass rate on the sampled subset is 82.4%, indicating that the proposed automatic annotation pipeline achieves practical annotation quality for large-scale corpus construction.

For turn-state annotations specifically, each turn is assigned one of four states: *complete*, indicating a semantically finished utterance; *incomplete*, indicating an unfinished or interrupted utterance; *backchannel*, representing brief acknowledgment or feedback; and *wait*, representing a request to pause or hold the conversation [19, 20]. Qwen3-32B [21] first predicts candidate turn states from textual context, and Qwen3-Omni [22] subsequently incorporates acoustic information for verification. On the expert-reviewed 20-hour subset, the automatically generated turn-state labels achieve 91% agreement with human annotations. When predictions disagree, the audio-based decision is preferred for incomplete turns and backchannels, where prosodic cues are particularly important.

Emotion, nonverbal events, gender, and age attributes are automatically annotated using Gemini 2.5 Pro [23]. Age attributes are categorized into five groups in DuplexConv: child, teenager, adult, middle-aged, and elderly. Paralinguistic annotations describe emotional states and nonverbal vocal events, including crying, coughing, throat clearing, sneezing, sighing, and other vocal behaviors beyond lexical content. SmoothConv adopts predefined emotion categories to facilitate annotation and quality control, including neutral, happy, confused, curious, angry, surprise, sad, contempt, worry, disgust, fear, and sorry. In contrast, DuplexConv uses free-form emotion descriptions to represent diverse emotional expressions.

2.2. Dataset Statistics

The released corpora contain Mandarin conversations from two scenarios: Tutoring and Social Chat. Each session includes two to four synchronized participant-level speech tracks, preserving natural speech overlap and long-range conversational context.

SmoothConv is a human-verified corpus containing 100 hours from 2,503 conversation clips. It covers natural two- to four-speaker conversations with diverse interaction patterns. Each conversation is annotated by four annotators and further reviewed by three senior annotators, covering transcripts, speaker identities, utterance boundaries, turn states, and interaction attributes. SmoothConv provides reliable supervision for fine-grained analysis and evaluation of multi-party speech interaction.

DuplexConv contains 2,000 hours from 93,709 clips constructed through the automatic annotation pipeline. The average clip duration is 76.8 s, ranging from 8 s to 618 s. It includes conversations with two to four speakers and provides scalable automatically generated annotations to support large-scale model training.

2.3. SmoothConv Benchmark

To facilitate reproducible evaluation, we further release the **SmoothConv Benchmark**, a task-oriented evaluation suite derived from the human-verified SmoothConv corpus. As shown in Table 2,

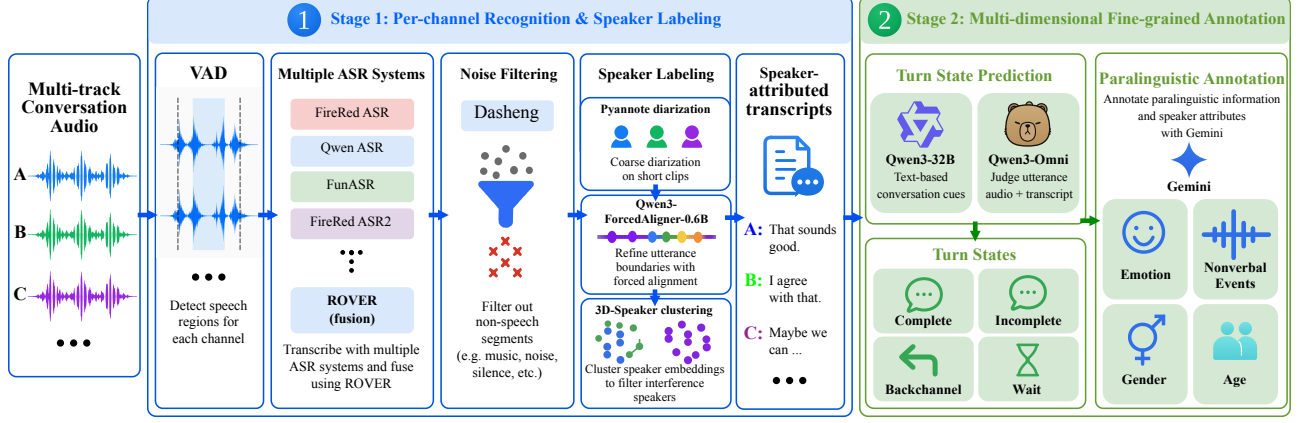


Fig. 1. Overview of the construction pipeline. Participant-level speech tracks are first processed for aligned transcription and speaker labeling, followed by multi-dimensional annotation of conversational behaviors and speaker-related information.

the benchmark provides dedicated evaluation subsets for different multi-party speech interaction tasks.

For speech separation and multi-speaker ASR (MSASR), we construct a shared evaluation subset containing synchronized participant-level speech tracks. The subset consists of natural two-to four-speaker conversations and follows a strict speaker-disjoint partition strategy, ensuring that no speaker identity appears in both training and evaluation sets. Moreover, the average speech overlap ratio is 3.33%, reflecting the sparse but challenging overlap patterns in real-world conversations.

Table 2. Statistics of the SmoothConv Benchmark subsets.

Speech Separation / MSASR Evaluation Set	
Duration	9.48 h
Clips	255
Average duration	133.9 s
Speaker configuration	2–4 speakers
Scenario	Tutoring 56.0% / Social 44.0%
Speaker split	Disjoint
Speech overlap	3.33%
Turn Detection Evaluation Set	
Turn State	Samples / Duration (h)
Complete	14709 / 9.64
Incomplete	3643 / 2.15
Backchannel	3080 / 0.42
Wait [†]	1000 / 0.71

[†] Synthesized samples generated only for evaluation.

For turn detection, we construct an independent evaluation subset consisting of short segmented dialogue clips with turn-state annotations. The subset mainly contains real-world dialogue segments with *complete*, *incomplete*, and *backchannel* states. Since explicit *wait* states are relatively sparse in natural conversations, we further introduce 1,000 synthesized *wait* samples. These samples are generated using DeepSeek V3 [24] for text generation and IndexTTS2 [25] for speech synthesis. All synthesized samples are manually checked and used only for evaluation.

3. EXPERIMENTS

We evaluate the proposed corpora on three downstream tasks closely related to multi-party speech interaction: speech separation, multi-speaker automatic speech recognition (MSASR), and conversational turn detection. These tasks cover three important aspects of interactive speech processing. Speech separation evaluates whether the corpus can support reliable speaker-level signal recovery, which serves as a prerequisite for subsequent speaker-aware interaction understanding. MSASR focuses on understanding “who said what”, providing the fundamental content representation required for interaction modeling. Turn detection further investigates conversational dynamics by modeling speaker transitions and interaction states.

All training and evaluation splits are strictly separated, with no shared conversation sessions or speakers. When SmoothConv is used for training, the corresponding evaluation samples are removed to avoid data leakage.

Table 3. Speech separation results under different training configurations. SI-SDRi (dB) and CER (%) are reported. DC (DuplexConv-derived mixtures), Sim. (simulated mixtures), and SC (SmoothConv) denote training sources. In-domain and OOD denote in-domain and out-of-domain evaluations.

Training data	In-domain (SI-SDRi↑/CER↓)		OOD (CER↓)	
	Simulated	SmoothConv	AliMeeting	AISHELL-4
Sim.	14.87 /31.99	13.55/81.24	65.29	68.49
DC	7.13/37.17	17.26 / 32.15	42.26	37.66
DC+Sim.	13.46/ <u>29.29</u>	15.63/45.41	53.67	40.75
DC+Sim.+SC	12.32/29.73	<u>15.70</u> / <u>42.59</u>	<u>45.48</u>	36.04

3.1. Speech Separation

For speech separation, we train an SPMamba [26] model using different training configurations based on simulated mixtures and DuplexConv-derived mixtures. All settings use 480 hours of train-

Table 4. MSASR results. DER, cpWER, and tcpWER are reported in percentage (%). Baseline denotes the original SoulX-Transcriber model. DC (DuplexConv) and DC+SC (DuplexConv + SmoothConv) denote adaptation settings.

Training Data	SmoothConv-Bench			AliMeeting-Near			AliMeeting-Far			AMI-SDM		
	DER↓	cpWER↓	tcpWER↓	DER↓	cpWER↓	tcpWER↓	DER↓	cpWER↓	tcpWER↓	DER↓	cpWER↓	tcpWER↓
Baseline	40.89	27.04	27.95	4.48	12.52	11.65	12.82	25.88	25.08	16.15	<u>73.66</u>	<u>97.75</u>
DC	<u>13.50</u>	<u>17.38</u>	<u>18.22</u>	<u>0.71</u>	<u>8.27</u>	<u>7.56</u>	20.29	30.17	30.92	<u>15.56</u>	74.04	97.56
DC+SC	12.73	16.87	17.62	0.68	8.14	7.43	<u>18.37</u>	<u>29.18</u>	<u>29.76</u>	14.57	73.28	97.31

ing data. The simulated mixtures are collected from WSJ0-2mix, WHAM! [27], WHAMR! [28], and Libri2Mix [29], while DuplexConv mixtures are constructed from participant-level conversational tracks. Hybrid settings combine the two types of mixtures. Evaluation is performed on simulated mixtures, the SmoothConv Benchmark, and two out-of-domain conversational datasets, AliMeeting [5] and AISHELL-4 [6]. SI-SDRi and CER are reported, where CER is computed using the X-ASR-zh-en model on separated outputs. For out-of-domain datasets, only CER is reported due to the lack of clean source references.

As shown in Table 3, simulated and conversational mixtures exhibit complementary advantages. DuplexConv-only training achieves competitive performance on conversational evaluation sets, while the hybrid setting provides a better trade-off between simulated and conversational conditions. These results demonstrate that participant-level speech tracks from natural conversations can effectively support multi-party speech separation.

3.2. Multi-speaker Automatic Speech Recognition

For MSASR, we adapt the SoulX-Transcriber [30] model using conversational data from DuplexConv and SmoothConv. LoRA [31] adaptation is applied while keeping the backbone model unchanged. All models are fine-tuned for one epoch with a learning rate of 1×10^{-5} , a warmup ratio of 0.05, and LoRA parameters of rank 8 and $\alpha = 32$. All training and evaluation splits are strictly separated. Evaluation is conducted on the SmoothConv Benchmark and external conversational datasets, including AliMeeting [5] and AMI-SDM [4]. DER, cpWER, and tcpWER are reported with a 0.25 s collar for diarization evaluation.

As shown in Table 4, DuplexConv adaptation substantially improves MSASR performance on the SmoothConv Benchmark, reducing both diarization and transcription errors compared with the baseline. Combining DuplexConv with SmoothConv further improves performance under several evaluation conditions, demonstrating the complementary benefits of scalable automatically annotated data and human-verified annotations. These results indicate that the proposed corpora effectively support speaker-aware content understanding for multi-party speech interaction. The performance degradation on far-field AliMeeting further suggests that acoustic mismatch remains a challenge for cross-domain conversational adaptation.

3.3. Turn Detection

Turn detection evaluates conversational behavior modeling, including speaker transitions and interaction states. We train FastTurn [20] using three training configurations: EasyTurn [19], DuplexConv

Table 5. Turn detection recall (%) under different training settings. ET (EasyTurn training set), DC (DuplexConv), and DC+SC (DuplexConv + SmoothConv) denote training configurations. C, I, B, and W denote Complete, Incomplete, Backchannel, and Wait states, respectively.

Training	EasyTurn-Test				SmoothConv-Bench			
	C	I	B	W	C	I	B	W
ET	93.33	85.00	<u>87.00</u>	98.00	78.29	<u>73.03</u>	92.57	95.70
DC	99.67	<u>87.67</u>	76.00	91.00	<u>87.67</u>	72.85	<u>93.78</u>	89.60
DC+SC	<u>98.00</u>	91.00	93.00	<u>96.00</u>	89.20	74.66	94.15	<u>91.30</u>

(DC), and DuplexConv combined with SmoothConv (DC+SC). Recall is reported for four interaction states: *complete*, *incomplete*, *backchannel*, and *wait*.

As shown in Table 5, different datasets provide complementary supervision for turn-state modeling. On EasyTurn-Test, the EasyTurn-trained model achieves the best performance on the *wait* state due to its dedicated annotations, while DuplexConv achieves stronger complete-turn recognition with large-scale conversational training data. Combining DuplexConv with SmoothConv further improves incomplete and backchannel detection, indicating that high-quality annotations help capture subtle interaction boundaries.

On the SmoothConv Benchmark, DC+SC achieves the best performance on most states. The improvements on incomplete and backchannel states demonstrate the value of combining large-scale conversational data with reliable annotations for modeling speaker coordination. The *wait* state remains challenging because explicit waiting behaviors are relatively rare in natural conversations, and the evaluation set contains synthesized samples, resulting in a potential gap from real interaction patterns.

4. CONCLUSION

In this work, we introduce SmoothConv and DuplexConv, two complementary Mandarin multi-party conversational corpora with participant-level tracks and multi-dimensional annotations. SmoothConv provides high-quality human-verified data for analysis and evaluation, while DuplexConv enables large-scale model training with automatically annotated conversations. We further release the SmoothConv Benchmark, a task-oriented evaluation suite covering different aspects of multi-party speech interaction. Experiments on speech separation, MSASR, and turn detection demonstrate the utility of these publicly available resources for modeling natural multi-party speech interaction.

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